FAST NUCES ISLAMABAD



Software Construction and Development

Assignment 02 — RAHBAR: Grounded RAG on Kubernetes

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Department:	**Software Engineering**

Assignment 02 — Submission Deliverables

Official Deliverables required per Section 8 of Assignment 02 Specification

1. GitHub Repository URL

- https://github.com/ali-akbar1214/rahbar-rag

  Public GitHub repository containing all source code, Kubernetes manifests, evaluation suites, and architectural documentation (synced at commit d211073).

2. Link to a Successful cd.yml Run

- https://github.com/ali-akbar1214/rahbar-rag/actions/runs/41708212822

  CD pipeline run on main branch: successfully tested unit suites, published SHA-tagged images to Docker Hub, deployed Kind ephemeral cluster, ran indexer Job, and verified Ingress routing and smoke tests.

3. Links to Images on Docker Hub (Showing SHA Tags)

- **API Image:** https://hub.docker.com/r/aliakbar1214/rahbar-api/tags

  Tag: d211073a30f171f943fbd3b96cb8f29f1ad970ac

- **Indexer Image:** https://hub.docker.com/r/aliakbar1214/rahbar-indexer/tags

  Tag: d211073a30f171f943fbd3b96cb8f29f1ad970ac

- **UI Image:** https://hub.docker.com/r/aliakbar1214/rahbar-ui/tags

  Tag: d211073a30f171f943fbd3b96cb8f29f1ad970ac

4. Committed Evaluation Files (eval/baseline.json and eval/results.json)

- **eval/baseline.json:** https://github.com/ali-akbar1214/rahbar-rag/blob/main/eval/baseline.json

  Committed baseline benchmark recording recall@3 (0.8125), recall@5 (0.8750), citation validity (1.0), and abstention correctness (1.0).

- **eval/results.json:** https://github.com/ali-akbar1214/rahbar-rag/blob/main/eval/results.json

  Committed evaluation results validating retrieval quality and honest abstention across all 16 golden test questions.